MATH 202 Your name: _____

Learning target T2, version 1

Use the Taylor series of familiar functions to find the Taylor series for a related function.

Consider the function $f(x) = \sin(x)$, whose Taylor series is

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

1. Write down a Taylor series for the function $\operatorname{sinc}(x) := \dfrac{\sin(x)}{x} = \dfrac{1}{x} \cdot \sin(x)$.
   (I am told that this function is important in signal processing for some reason.)

2. If $g(x) = \operatorname{sinc}(x)$, find the Taylor series for an antiderivative $G(x)$ such that $G(0) = 0$.
   (This is called the "sine integral" $\operatorname{Si}(x)$ and it is also apparently important in signal processing.)

MATH 202 Your name: _____

Learning target T2, version 2

Use the Taylor series of familiar functions to find the Taylor series for a related function.

Consider the function $f(x) = \sin(x)$, whose Taylor series is

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

1. Write down a Taylor series for the function $\sin(x^2)$.

2. The Fresnel integral $S(x) = \int \sin(x^2)\, dx$ is important in optics, but as you may surmise, there is not an elementary antiderivative of $\sin(x^2)$. Darn! Guess we'll have to write down a power series representation for it instead.